



interfaceNode - *Manual*

Development 1.0.2

Hardware Revision: B.0.x

The Atkin Engineering **interfaceNode** is a Wireless RS-485 bridge that provides bridging connections between RS-485, USB, TCP sockets, and WebSockets for various control devices.

front

rear



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1. Connections

Front

- USB Type-C, USB 2.0 full-speed port [[^]usbpower]
- 1x Status LED
- 1x System button

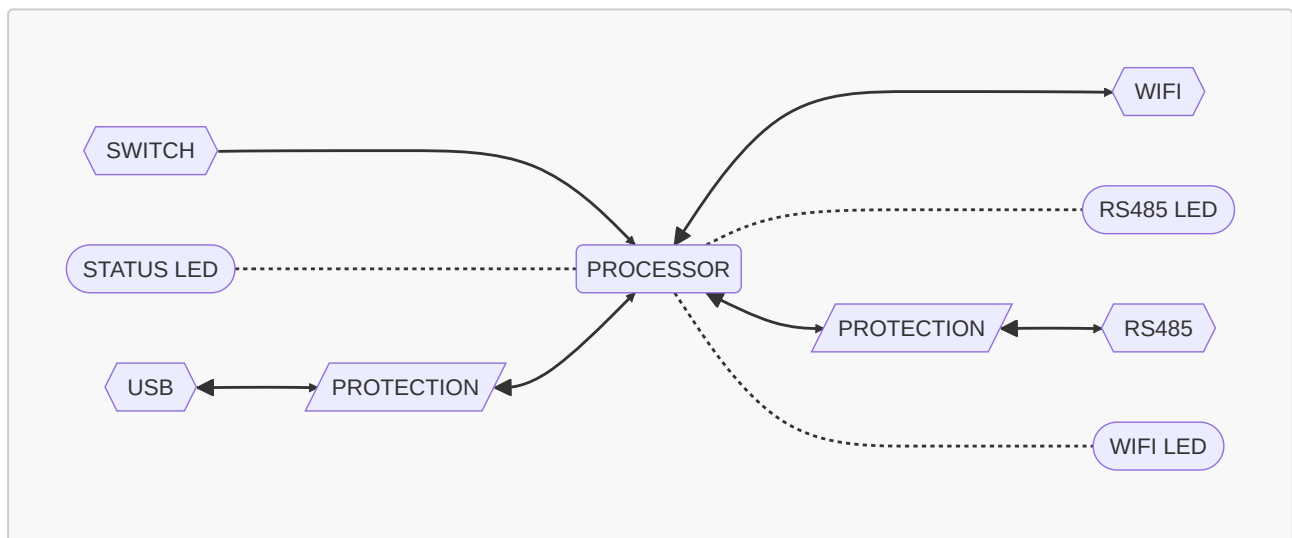
Rear

- 2x RS-485 (parallel RJ12 6P6 ports) supplies power & data [[^]rs485power]
- 2x RGB status LEDs
- Full / Half duplex (hardware jumper - see [[^]jumpers])
- Termination **None** / **120ohm** (hardware jumper - see [[^]jumpers])

RS-485 pinout (RJ12 6P6)

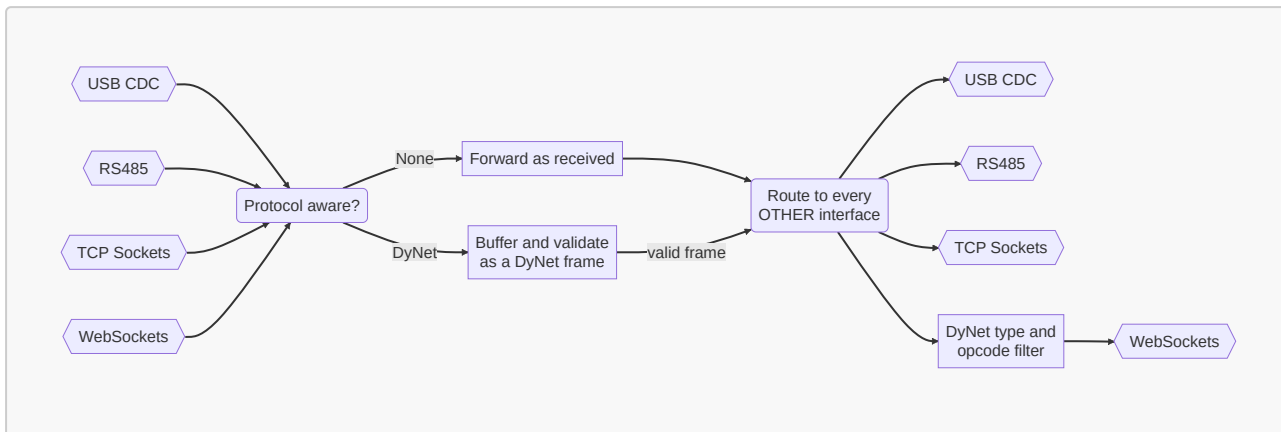
1. Ground
2. Ground
3. Data +
4. Data -
5. 12/24 VDC
6. 12/24 VDC

2. Block diagram



3. Operation flow

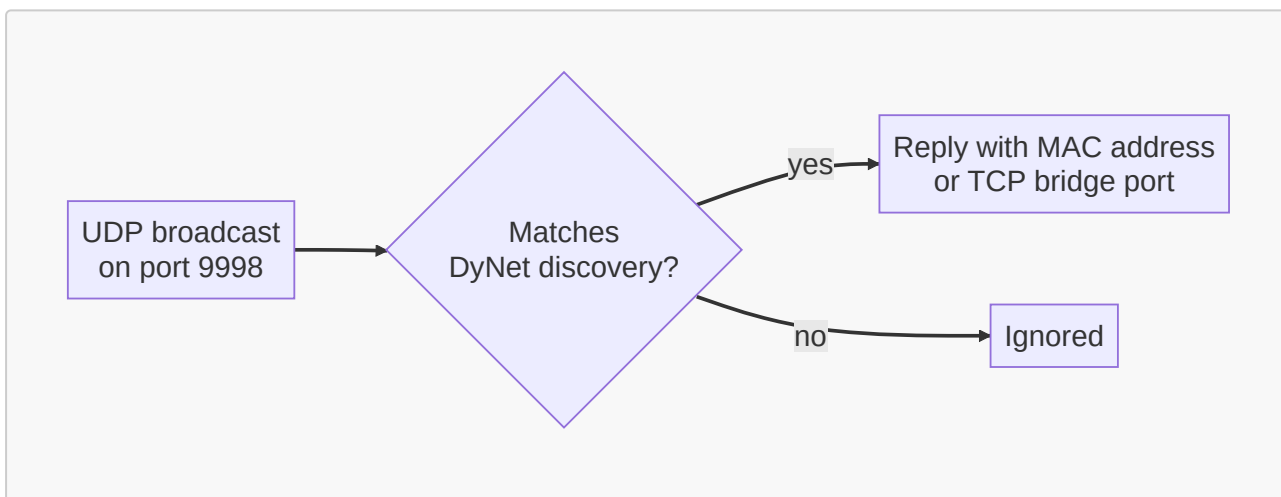
Data arriving on USB, RS485, a TCP socket or a WebSocket is handled the same way, whichever interface it came in on:



A frame is never sent back out of the interface it arrived on. With **Protocol aware** set to **DyNet**, incomplete data is held until the rest of the frame arrives, and data that fails its checksum is discarded while the stream re-synchronises on the next valid frame. With it set to **None**, data is passed straight through with no validation.

Only WebSocket delivery is filtered - see [WebSocket DyNet filtering](#). USB, RS485 and TCP always receive every frame.

The DyNet discovery beacon (see [Sockets](#)) is a separate process. It does not pass through the flow above, and nothing it receives or sends is routed to any bridge interface:



4. Default mode

A factory-reset **interfaceNode** boots in **Access point** mode with:

- SSID: **interfaceNodeXXYY**, where **XXYY** is the device's serial number (the last two bytes of its MAC address, in hex - see [Device](#))
- Password: **password**
- Security: **WPA2/WPA3 PSK**
- IP address: **192.168.4.1**, with a DHCP server for connected devices
- **Protocol aware**: **DyNet**
- TCP port: **50000**
- RS485 baud rate: **9600**
- WebSocket DyNet filtering: DyNet 1 forwarded with opcodes **0-255**; DyNet 2 not forwarded (see [WebSocket DyNet filtering](#))

Note: settings are also restored to these defaults after every firmware update - see [Firmware](#).

5. System button

The system button supports two gestures:

- **Short press** Switches between operating mode and configuration mode: the device reboots into configuration mode if it is operating, or back into operating mode if it is already in configuration mode. See [System](#) for what configuration mode means.
- **2 second hold** Resets all settings to factory defaults and reboots into [default mode](#).

Note: the button is not active immediately at power-on - it responds once the rest of the system (network, bridge or web interface, beacon) has started. It also stops responding once a reboot has been triggered by any means (the button itself, or an action in the web interface); a press during the short delay before the restart is ignored.

6. Config mode

Configuration mode runs the web interface used to view and change the device's settings. See [System](#) for how it relates to normal bridge operation, and [Default mode](#) / [System button](#) for how to enter and leave it.

Open the device in a web browser at its IP address ([192.168.4.1](#) in [default mode](#)), or by its mDNS hostname, e.g. [interfacenode3080.local](#) for serial number [3080](#). mDNS works on most networks and operating systems without any setup.

The page is divided into the sections below. Only the settings that apply are shown: the access point or station settings for the selected WiFi mode, and the DyNet filters only when [DyNet](#) is selected. Hardware-only settings (RS485 termination and duplex - see [Connections](#)) are not part of the web interface.

Device

Read-only details: status, serial number, hostname, MAC address, firmware version, build date, SDK version and the running firmware partition.

Network

Setting	Values
WiFi mode	Access point / Station - see Operation mode
Access point Network name	Text
Access point Password	Text, 8+ characters
Access point Channel	1-13
Access point Security	Open, WEP, WPA PSK, WPA2 PSK, WPA/WPA2 PSK, WPA3 PSK, WPA2/WPA3 PSK
Access point Device IP address	Default (192.168.4.1) / Custom (IP address, network mask, gateway, DNS)
Station Network name	Text
Station Security	As access point, plus WPA2 Enterprise and WPA3 Enterprise
Station Password	Text, 8+ characters
Station Enterprise sign-in	Username, identity, server certificate validation and certificate upload (Enterprise security only)
Station Device IP address	Automatic (DHCP) (default) / Static (IP address, network mask, gateway, DNS)

Protocol

Setting	Values
Protocol awareness	None / DyNet - see Operation flow
DyNet 1 opcodes	List of opcodes sent to WebSocket clients, e.g. 0-255 - see WebSocket DyNet filtering
DyNet 2 opcodes	As above, for DyNet 2

Bridge

Setting	Values
TCP port	49152-65535 - see Sockets
RS-485 baud rate	9600 / 19200 / 38400 / 57600 / 115200 - see Serial

Saving and maintenance

Action	Effect
Save and reboot	Stores every setting and reboots into operating mode
Reboot	Restarts the device without changing any settings
Factory reset	Restores all settings to their factory defaults, saves them, and reboots
Install firmware	Updates the firmware - see Firmware

See [System](#) for details.

7. Operation mode

interfaceNode can operate in two WiFi configurations:

1. **Access point** Creates a WiFi network for other devices to connect to.
2. **Station** Connects to an existing WiFi network as a client.

The **Protocol awareness** setting controls how the bridge handles data:

- **None** Data is forwarded between interfaces as received, with no validation.
- **DyNet** Data is buffered and validated as DyNet 1 or DyNet 2 frames (checksum-checked). Invalid data is discarded and the stream re-synchronises automatically.

Access point

- **Network name** SSID of the access point.
- **Password** Password for the access point.
Note: must be at least 8 characters.
- **Channel** WiFi channel (1-13).
- **Security** Security type: **Open**, **WEP**, **WPA PSK**, **WPA2 PSK**, **WPA/WPA2 PSK**, **WPA3 PSK** or **WPA2/WPA3 PSK**. **WPA2/WPA3 PSK** (the default) is recommended.

IP address settings:

- **Default** **192.168.4.1**, with the built-in DHCP server.
- **Custom** Set your own IP address, network mask, gateway and DNS server. Leaving gateway or DNS blank offers none to connected devices.

Note: configuration mode always uses the default address, so the device can always be reached for setup even if a custom address is wrong.

Station

- **Network name** SSID of the network to join.
- **Security** Security type of the network being joined: **Open**, **WEP**, **WPA PSK**, **WPA2 PSK**, **WPA/WPA2 PSK**, **WPA3 PSK**, **WPA2/WPA3 PSK**, **WPA2 Enterprise** or **WPA3 Enterprise**.
- **Password** Password of the network being joined. Must be at least 8 characters. For Enterprise networks this is the sign-in password.

If **WPA2 Enterprise** or **WPA3 Enterprise** is selected, an **Enterprise sign-in (EAP)** group appears:

- **Username** EAP username.
- **Identity** EAP identity string.
- **Validate server certificate / Don't validate** Whether to check the RADIUS server's certificate.
- **RADIUS / CA certificate** File picker and **Upload certificate** button. Upload the RADIUS server's CA certificate (PEM format) to enable validation.
Note: the certificate is used from the *next* WiFi connection attempt. If validation is selected but no certificate has been uploaded, the connection proceeds without validation.

PEAP, TTLS and EAP-FAST are supported; the RADIUS server selects the method.

IP address settings:

- **Automatic (DHCP)** Network-assigned IP address (default).
- **Static** Manually set IP address, network mask, gateway and DNS server.

8. Sockets

interfaceNode runs a TCP bridge server that multiple clients can connect to over the network, in either Access Point or Station mode.

- **TCP port** Listening port. Must be between 49152 and 65535.

Every TCP client is bridged the same way as USB and RS485 - see [Operation flow](#). Data received from one TCP client is forwarded to USB, RS485, WebSocket clients and every *other* TCP client, never back to the client that sent it. Up to 5 TCP clients can be connected at once. A further client is disconnected as soon as it connects.

The device also advertises itself by mDNS, including its TCP and WebSocket ports, so compatible software can find it on the network.

WebSocket bridge

interfaceNode also runs a dedicated WebSocket server at `ws://<device address>:8080/ws`, separate from the TCP bridge and from the configuration web interface. It runs in operating mode only. Up to 5 WebSocket clients can connect at once, and each is bridged the same way as USB, RS485 and TCP. While 5 are connected, a further client's connection attempt times out.

Data is exchanged as binary WebSocket messages. Clients should send binary messages: the device forwards the bytes of any message it receives, so a text message would put its character codes on the bus rather than the intended bytes.

WebSocket DyNet filtering

With **Protocol awareness** set to **DyNet**, the DyNet frames sent to WebSocket clients can be limited, for example to reduce traffic to a browser-based display. USB, RS485 and TCP are not affected.

Two rules apply to each DyNet type (DyNet 1 and DyNet 2):

1. **Forwarding** - whether that frame type is sent to WebSocket clients at all. By default DyNet 1 is forwarded and DyNet 2 is not. These two switches are not yet in the web interface.
2. **Opcode list** - which opcodes of that type are sent. Only frames whose opcode is listed are forwarded; all others are dropped. The opcode is byte 3 of a DyNet 1 frame and byte 2 of a DyNet 2 frame.

Opcode lists are set in the web interface under **Protocol**. A list is one or more values from 0 to 255, separated by commas, with a dash for a range:

List	Opcodes sent
0-255	All
1-255	All except opcode 0
1, 5, 20-30	1, 5, and 20 to 30

The defaults are 0-255 for DyNet 1 and 1-2 for DyNet 2. An invalid list is rejected when saving, with a message saying what is wrong.

Data that is not a DyNet frame, and all data when **Protocol awareness** is **None**, is always sent to WebSocket clients.

DyNet discovery beacon

When **Protocol awareness** is set to **DyNet** and the device is operating (not in configuration mode), **interfaceNode** listens for DyNet discovery broadcasts on UDP port 9998 and replies with its MAC address and TCP bridge port.

Note: this is a separate listener - it does not go through the bridge, and discovery traffic is never forwarded to or from any bridge interface. See [Operation flow](#).

9. Serial

- **RS-485 baud rate** 9600, 19200, 38400, 57600 or 115200.

RS485 duplex mode (full or half) and termination are set with hardware jumpers - see [Connections](#). RS485 is bridged to USB, TCP and WebSocket clients the same way as any other interface - see [Operation flow](#).

10. Led status

Rebooting (all LEDs, momentarily, immediately before the device restarts):

- **White** Rebooting - shown whatever triggered the restart (saving settings, factory reset, a firmware update, or the system button).

Status LED:

- **Green** Starting up.
- **Yellow** Configuration mode - stays until the device is rebooted out of it.
- **Magenta** Operating in Access point mode.
- **Blue** Operating in Station mode.

RS485 LED

- **Slow red pulse** Idle - alternates between bright and dim red every 2 seconds.
- **Fast red flicker** RS485 data activity - returns to idle 0.8 seconds after the last data.

WiFi LED

- **Slow blue pulse** No TCP client connected.
- **Solid dim blue** At least one TCP client connected, idle.
- **Fast blue flicker** TCP data activity.

Note: the WiFi LED shows TCP connections and data only. The WiFi mode is shown on the status LED.

11. System

Important: the web configuration interface is only available in configuration mode. In normal operation (the bridge running), it is not available, and the bridge doesn't run in configuration mode. Switching between the two is a reboot - see [System button](#).

- **Save and reboot** When any setting has been changed, a bar appears at the bottom of the page with this button. It stores every setting and reboots into operating mode. If the network settings changed, reconnect to the device at its new address afterwards.
- **Reboot** Restarts the device without changing any settings. Unsaved changes are lost.
- **Factory reset** Restores all settings, including WiFi, to their factory defaults and reboots into [default mode](#).

Each action asks for confirmation before it runs. The browser also warns before leaving the page with unsaved changes.

System log

Recent log output is viewable at [/systemlog.html](#), colour-coded by severity (error, warning, info, debug, verbose).

Like the rest of the web interface, it is only available in configuration mode.

The log is kept in a fixed-size, 128 KB in-memory buffer. When full, it clears and starts again.

The log survives a reboot triggered by the device itself (**Save and reboot**, **Factory reset**, **Reboot**, or the system button) and continues where it left off. It is cleared by a power cycle or a firmware update, which are logged as "STARTING COLD BOOT"; any other restart logs "STARTING WARM BOOT" and keeps the existing entries.

12. Firmware

- **Firmware file (.bin)** Choose the firmware file.
- **Install firmware** Confirm, then upload the firmware to **interfaceNode**.

Once the browser has sent the file, the page shows "Writing to device flash - keep the device powered and connected" while the device writes it, which can take several seconds. Do not disconnect or power off the device during this time. The device then restarts with the new firmware.

Important: with the current firmware, all settings, including WiFi, are restored to their defaults after a firmware update. The device restarts in **default mode**, so note your settings before updating and re-enter them afterwards.

If the new firmware fails to start, the device automatically returns to the previous firmware.

Firmware files @ <https://github.com/AtkinEngineering/interfaceNode/releases>

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[^usbpower]: USB Type-C port does not supply power.

[^rs485power]: RS-485 (parallel RJ12 6P6 ports) supplies power & data to **interfaceNode**.

[^jumpers]: Termination and duplex are both set via jumpers internal to the enclosure, not configurable from the web interface - changing either requires opening the unit. For duplex specifically: the jumper selects between the receiver always being enabled (full duplex) or tied to the transmit-enable line (half duplex). The firmware drives a single direction-control signal in either case - the jumper, not the firmware, determines which duplex mode results.

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